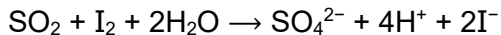
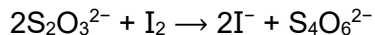


- 1 Sulfur dioxide is a preservative that can be found in small amount in wines. The sulfur dioxide content in wine can be determined through back titration.

A 25 cm³ sample of wine is reacted with 30.0 cm³ of 0.01 mol dm⁻³ of excess aqueous iodine, oxidising the sulfur dioxide in the wine to sulfate, SO₄²⁻, in the process.



The unreacted iodine requires exactly 24.80 cm³ of 0.02 mol dm⁻³ sodium thiosulfate for complete reaction.



What is the concentration of sulfur dioxide, in mol dm⁻³, in the wine?

- A** 2.08×10^{-3} **B** 2.48×10^{-3} **C** 9.92×10^{-3} **D** 2.77×10^{-2}